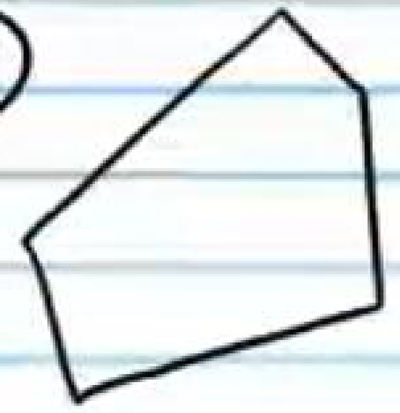
Name
Date
Course

Special Quadrilaterals (Part I) and Angles of Polygons (Homework)

Find the sum of the measures of the interior angles of the following convex polygons. Show your work.

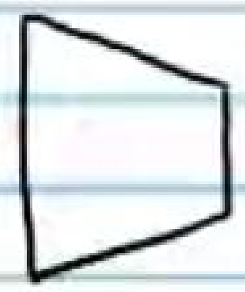
Answers

① 

540°

② 19-gon — 3,060°

③ Hexagon — 720°

④ 

360°

⑤ Nonagon — 1260°

⑥ 

720°

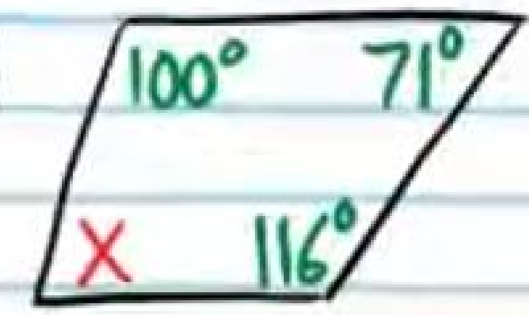
If each number below is the sum of the measures of the interior angles of a convex polygon, find the number of sides of each polygon. Show your work.

⑦ 2160° — 14

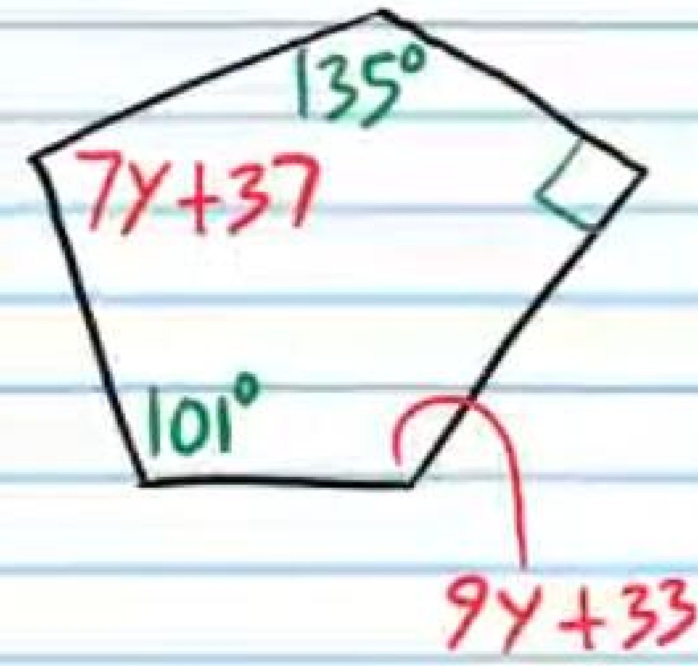
⑧ 3600° — 22

⑨ 4,320° — 26

Find the unknowns in each polygon below.

⑩ 

X = 73°

⑪ 

Y = 9

Find the sum of the measures of the exterior angles of the following convex polygons.

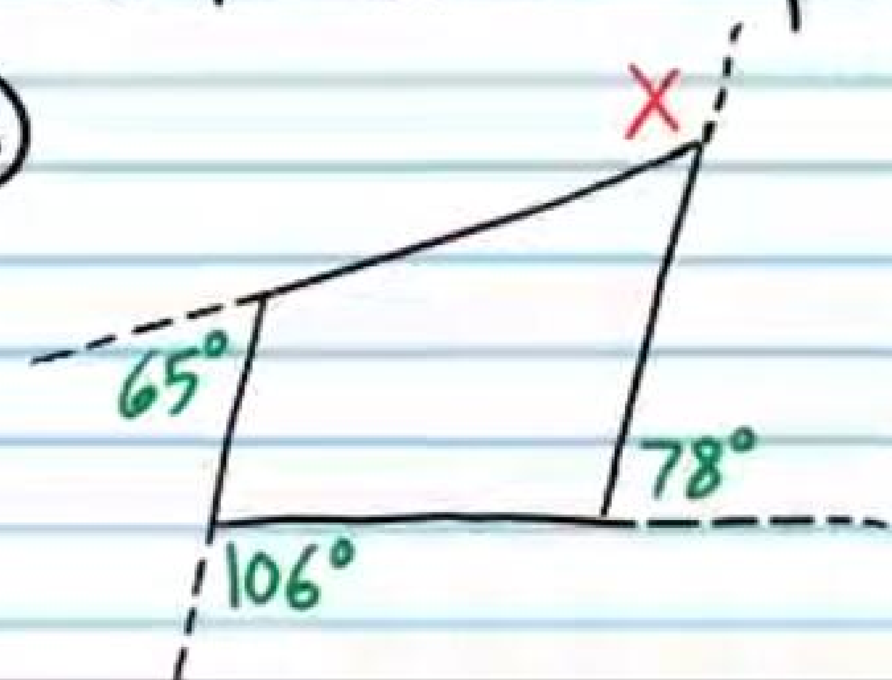
⑫ Heptagon — 360°

⑬ Decagon — 360°

⑭ 100-gon — 360°

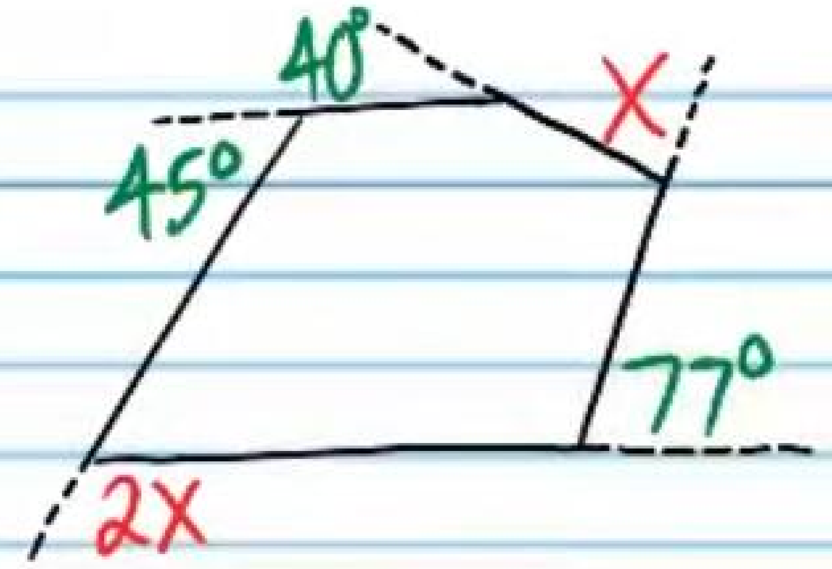
⑮ Nonagon — 360°

Find X in each problem below.

⑯ 

X = 111°

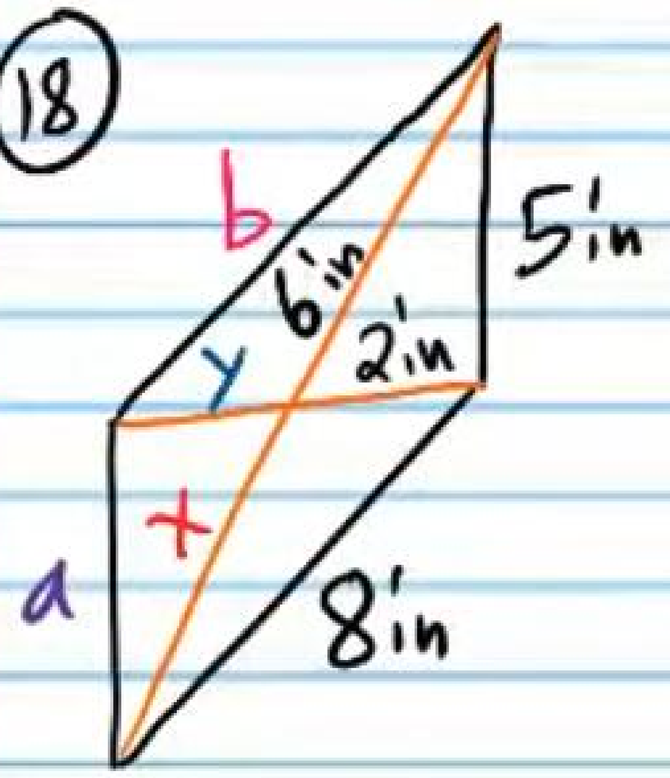
(17)



$\boxed{X = 66°}$

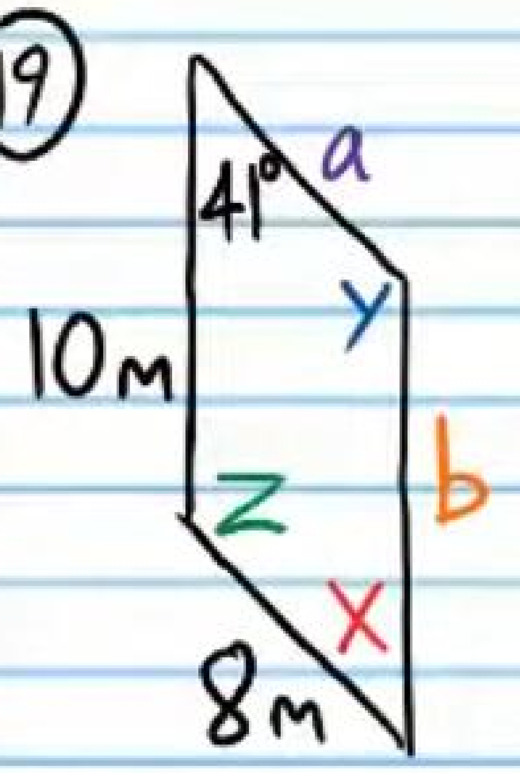
Find the unknown values in the parallelograms below.

(18)



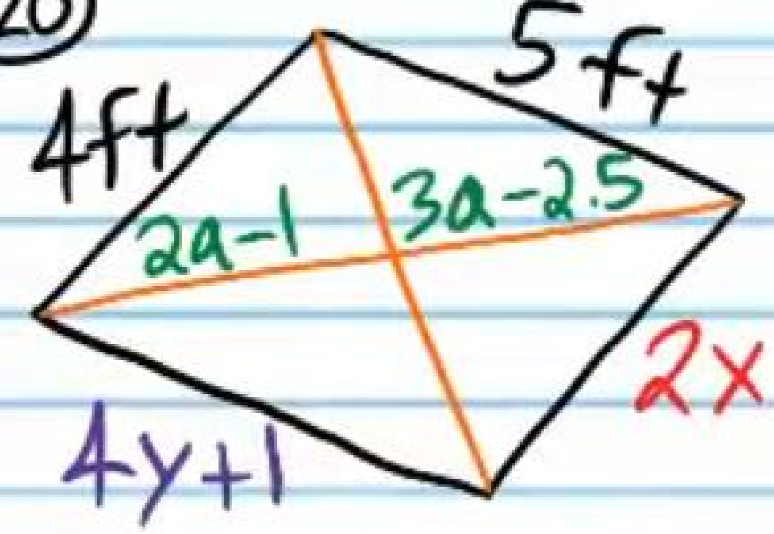
$\boxed{\begin{array}{l} a = 5 \text{ in} \\ b = 8 \text{ in} \\ X = 6 \text{ in} \\ y = 2 \text{ in} \end{array}}$

(19)



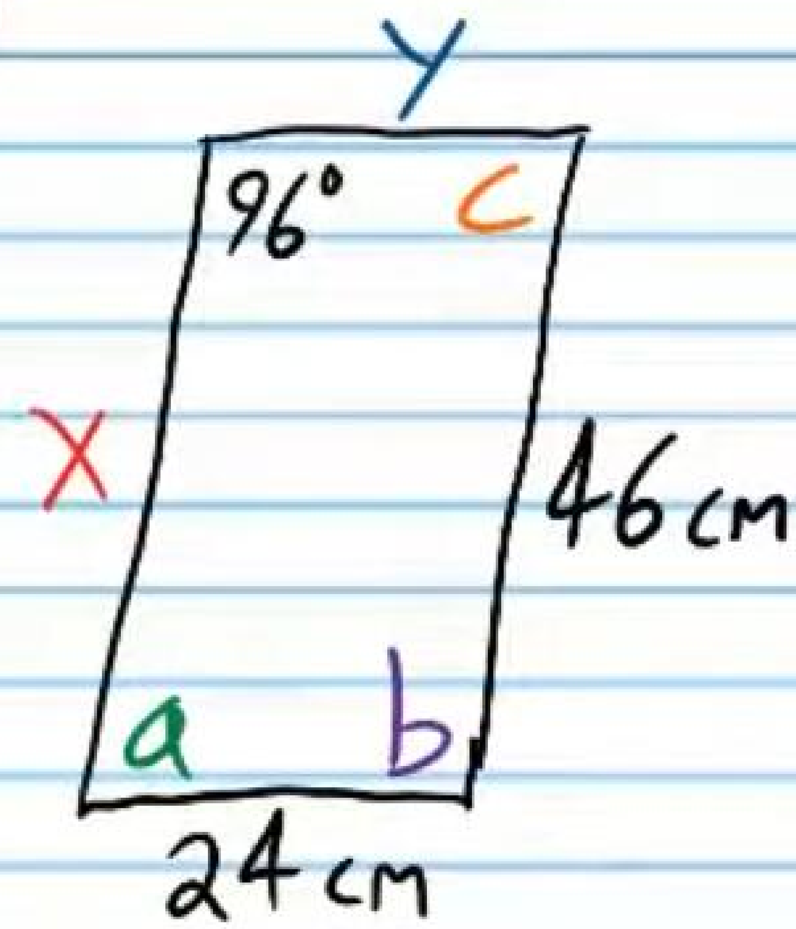
$\boxed{\begin{array}{l} X = 41° \\ y = 139° \\ Z = 139° \\ a = 8 \text{ m} \\ b = 10 \text{ m} \end{array}}$

(20)



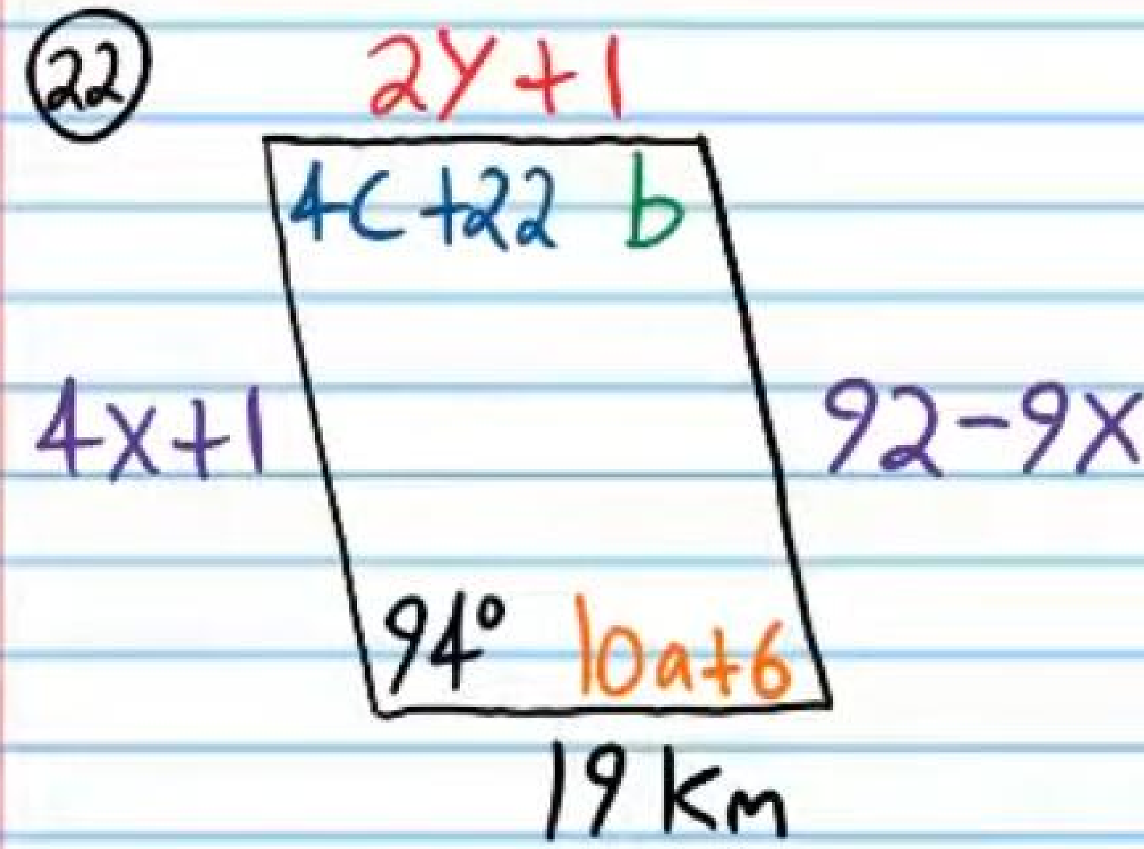
$\boxed{\begin{array}{l} X = 2 \\ y = 1 \\ a = 1.5 \end{array}}$

(21)



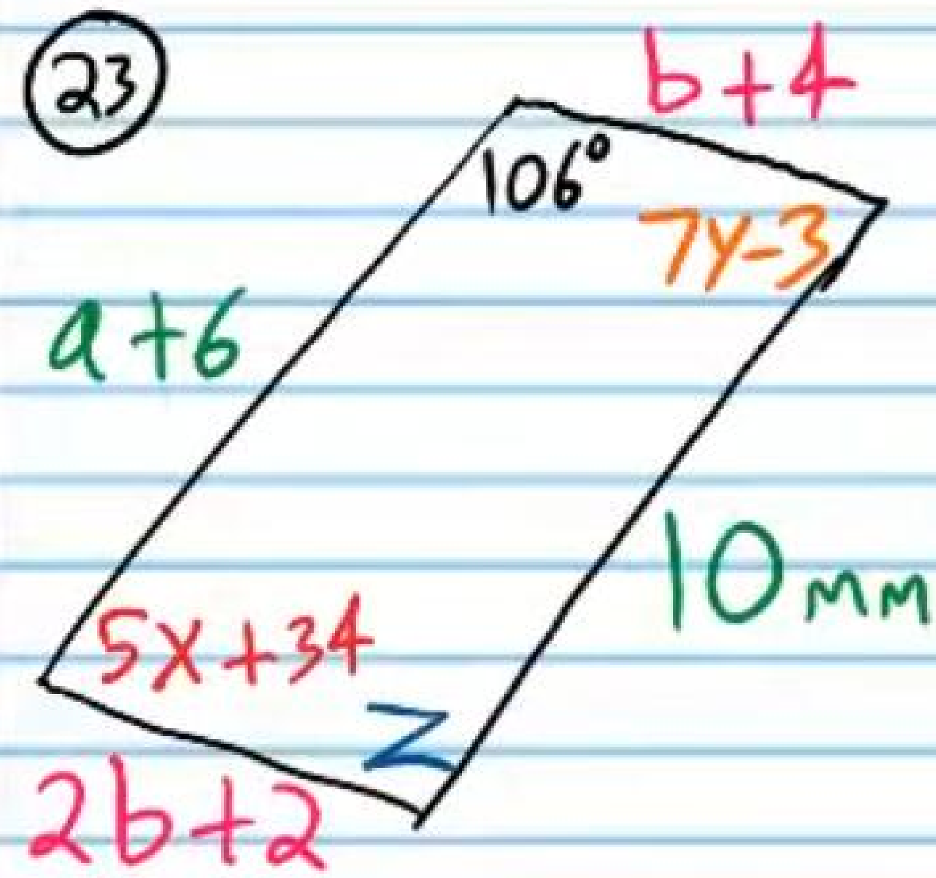
$\boxed{\begin{array}{l} y = 24 \text{ cm} \\ X = 46 \text{ cm} \\ a = 84° \\ b = 96° \\ c = 84° \end{array}}$

(22)



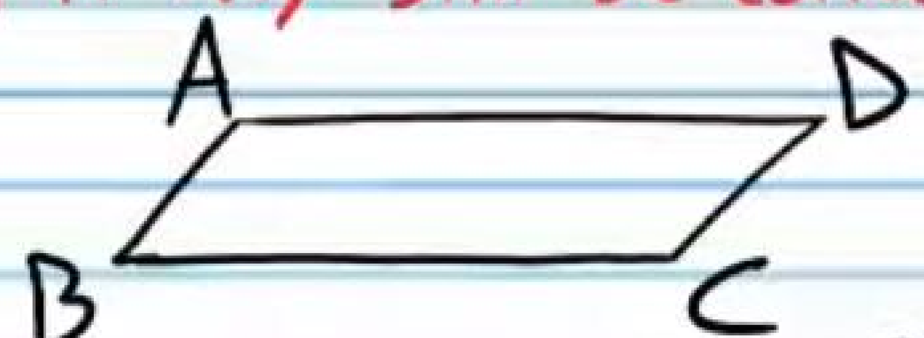
$\boxed{\begin{array}{l} y = 9 \\ X = 7 \\ a = 8 \\ b = 94° \\ c = 16 \end{array}}$

(23)



$\boxed{\begin{array}{l} X = 8 \\ y = 11 \\ Z = 106° \\ a = 4 \\ b = 2 \end{array}}$

★ For the problems below, use the following pre-labeled parallelogram. For some problems, there is more than one way to prove the statement, so if your answer does not match exactly, it may still be correct.



(24) Prove the converse of the Parallelogram Opposite Sides Theorem, which is written in if-then form below. Hint: You must draw an auxiliary line, and you must use the Alternate Interior Angles Converse Theorem.

If opposite sides of a quadrilateral are congruent, then the quadrilateral is a parallelogram.

(25) Prove the converse of the Parallelogram Consecutive Angles Theorem, which is written in if-then form below. Hint: You must use the Consecutive Interior Angles Converse Theorem.

If consecutive angles of a quadrilateral are supplementary, then the quadrilateral is a parallelogram.

(26) Prove the converse of the Parallelogram Opposite Angles Theorem, which is written in if-then form below. Hint: You must use the Polygon Interior Angles Theorem and the Consecutive Interior Angles Converse Theorem.

If opposite angles of a quadrilateral are congruent, then the quadrilateral is a parallelogram.

(27) Prove the converse of the Parallelogram Diagonals Theorem, which is written in if-then form below. Let E be the point of bisection.

If the diagonals of a quadrilateral bisect each other, then the quadrilateral is a parallelogram.

Answers for #'s 24-27

(24)

Statement	Reason
I) $\overline{AB} \cong \overline{DC}$ & $\overline{AD} \cong \overline{BC}$.	Given.
II) $\overline{AC} \cong \overline{AC}$	Ref. Prop of $\cong$.
III) $\triangle ABD \cong \triangle CDB$	SSS
IV) $\angle BAC \cong \angle DCA$	CPCTC
V) $\overline{AB} \parallel \overline{DC}$.	AIACT
VI) $\angle DAC \cong \angle BCA$.	CPCTC
VII) $\overline{AD} \parallel \overline{BC}$.	AIACT
VIII) ABCD is a parallelogram.	Def. of ▱.

(25)

Statement	Reason
I) $\angle A$ and $\angle B$ are supplementary.	Given.
II) $\overline{AD} \parallel \overline{BC}$.	CIACT
III) $\angle A$ and $\angle D$ are supplementary.	Given.
IV) $\overline{AB} \parallel \overline{DC}$.	CIACT
V) ABCD is a parallelogram.	Def. of ▱

(26)

Statement	Reason
I) $\angle A \cong \angle C$ & $\angle B \cong \angle D$.	Given.
II) $m\angle A + m\angle B + m\angle C + m\angle D = 360$	PIAT
III) $m\angle A + m\angle B + m\angle A + m\angle B = 360$.	Subs. Prop of =.
IV) $2m\angle A + 2m\angle B = 360$	simplify.
V) $m\angle A + m\angle B = 180$	Div. Prop of =.
VI) $\angle A$ is supplementary to $\angle B$.	Def. of Supp. $\angle$'s.
VII) $\overline{AD} \parallel \overline{BC}$.	CIACT
From II. VIII) $m\angle A + m\angle D + m\angle A + m\angle D = 360$.	Subs. Prop of =.
IX) $2m\angle A + 2m\angle D = 360$.	simplify.
X) $m\angle A + m\angle D = 180$	Div. Prop of =.
XI) $\angle A$ is supplementary to $\angle D$.	Def. of Supp. $\angle$'s.
XII) $\overline{AB} \parallel \overline{DC}$.	CIACT
XIII) ABCD is a parallelogram.	Def. of ▱.

(27)

Statement	Reason
I) $\overline{AC}$ bisects $\overline{BD}$.	Given.
II) $\overline{AE} \cong \overline{EC}$ & $\overline{BE} \cong \overline{ED}$.	Def. of Seg. bisector.
III) $m\angle AEB = m\angle CED$.	Vert. $\angle$'s Thm.
IV) $\triangle AEB \cong \triangle CED$	SAS
V) $\angle BAE \cong \angle DCE$	CPCTC
VI) $\overline{AB} \parallel \overline{DC}$	AIACT
VII) $m\angle AED = m\angle CEB$	Vert. $\angle$'s Thm.
VIII) $\triangle AED \cong \triangle CEB$	SAS
IX) $\angle DAE \cong \angle BCE$	CPCTC
X) $\overline{AD} \parallel \overline{BC}$	AIACT
XI) ABCD is a parallelogram.	Def. of ▱